

## MARGATE, South Africa

WMO: 685910

Lat: 30.859S

Lon: 30.343E

Elev: 154

StdP: 99.49

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 11.4       | 12.4 | -1.0                          | 3.5 | 17.6 | 1.1 | 4.2 | 17.5 | 10.4                  | 16.6 | 9.3 | 18.2 | 3.6                   | 280  | 0.332 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 4.8                    | 28.6            | 22.7 | 27.7 | 22.8 | 27.0 | 22.7 | 25.1                | 27.0 | 24.5 | 26.4 | 23.9 | 25.9 | 4.3                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.5                            | 19.8 | 26.2 | 23.9 | 19.1 | 25.7 | 23.2 | 18.3 | 25.3 | 77.5          | 27.1 | 75.0 | 26.4 | 72.7 | 25.9 | 27.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |     |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|-----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |     |
| (4) 9.8           | 8.6  | 7.7 | DB                         | 9.2 | 32.0               | 3.8 | 1.3                                                | 6.5 | 32.9       | 4.2 | 33.7       | 2.1 | 34.4       | -0.7 | 35.4 | (4) |
| (5)               |      |     | WB                         | 5.2 | 25.7               | 2.7 | 0.8                                                | 3.3 | 26.3       | 1.7 | 26.8       | 0.1 | 27.2       | -1.9 | 27.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 20.5        | 23.2  | 23.8  | 23.0  | 21.3  | 19.9  | 18.3  | 17.9  | 18.3  | 18.5  | 19.2  | 20.3  | 22.1  | (6)  |  |
|                                                                        | DBStd       | 2.92        | 1.86  | 1.82  | 1.91  | 2.10  | 2.16  | 1.94  | 2.18  | 2.21  | 2.25  | 2.27  | 2.26  | 2.03  | (7)  |  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 147         | 1     | 1     | 1     | 3     | 9     | 22    | 33    | 28    | 24    | 17    | 8     | 1     | (9)  |  |
|                                                                        | CDD10.0     | 3824        | 408   | 387   | 403   | 339   | 308   | 250   | 244   | 257   | 255   | 286   | 309   | 376   | (10) |  |
|                                                                        | CDD18.3     | 928         | 150   | 153   | 146   | 93    | 59    | 22    | 19    | 26    | 29    | 45    | 67    | 119   | (11) |  |
|                                                                        | CDH23.3     | 3023        | 641   | 759   | 554   | 203   | 116   | 44    | 48    | 61    | 42    | 57    | 126   | 372   | (12) |  |
| (13)                                                                   | CDH26.7     | 236         | 44    | 63    | 36    | 13    | 16    | 6     | 7     | 8     | 7     | 6     | 7     | 23    | (13) |  |
| (14)                                                                   | Wind        | WSAvg       | 3.6   | 3.8   | 3.7   | 3.4   | 3.5   | 3.0   | 3.2   | 3.7   | 3.9   | 4.2   | 3.9   | 3.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1035        | 121   | 91    | 137   | 72    | 38    | 38    | 49    | 41    | 80    | 119   | 134   | 139   | (15) |  |
|                                                                        | PrecMax     | 1660        | 352   | 213   | 331   | 179   | 120   | 340   | 264   | 145   | 280   | 288   | 352   | 316   | (16) |  |
|                                                                        | PrecMin     | 542         | 45    | 14    | 39    | 4     | 0     | 1     | 0     | 4     | 11    | 19    | 34    | 49    | (17) |  |
|                                                                        | PrecStd     | 268         | 64    | 47    | 86    | 47    | 34    | 70    | 67    | 36    | 55    | 64    | 77    | 60    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 29.1  | 29.5  | 28.7  | 28.1  | 28.6  | 27.1  | 27.4  | 28.0  | 27.4  | 26.8  | 27.6  | 28.8  | (19) |  |
|                                                                        |             | MCWB        | 24.4  | 24.2  | 23.9  | 22.1  | 18.2  | 15.9  | 15.7  | 16.6  | 18.8  | 20.7  | 22.8  | 23.8  | (20) |  |
|                                                                        | 2%          | DB          | 27.9  | 28.2  | 27.7  | 26.4  | 25.9  | 24.5  | 24.5  | 25.0  | 24.2  | 24.6  | 25.8  | 27.1  | (21) |  |
|                                                                        |             | MCWB        | 23.9  | 23.8  | 23.6  | 21.6  | 18.4  | 15.4  | 15.3  | 16.5  | 19.1  | 20.4  | 22.4  | 23.3  | (22) |  |
|                                                                        | 5%          | DB          | 27.0  | 27.4  | 26.9  | 25.4  | 24.4  | 22.9  | 22.5  | 23.0  | 22.8  | 23.5  | 24.7  | 26.1  | (23) |  |
|                                                                        |             | MCWB        | 23.4  | 23.5  | 23.2  | 21.2  | 18.2  | 15.5  | 15.2  | 16.7  | 18.7  | 19.9  | 21.6  | 22.7  | (24) |  |
|                                                                        | 10%         | DB          | 26.2  | 26.7  | 26.0  | 24.4  | 23.2  | 21.7  | 21.2  | 21.7  | 21.7  | 22.5  | 23.8  | 25.2  | (25) |  |
|                                                                        |             | MCWB        | 22.8  | 23.2  | 22.7  | 20.8  | 18.2  | 15.8  | 15.2  | 16.8  | 18.2  | 19.4  | 20.9  | 22.1  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 26.0  | 25.6  | 25.6  | 23.6  | 21.8  | 20.1  | 19.0  | 20.2  | 21.3  | 22.6  | 24.2  | 25.5  | (27) |  |
|                                                                        |             | MCDB        | 27.5  | 27.5  | 27.5  | 26.1  | 24.2  | 22.1  | 22.0  | 23.3  | 23.7  | 24.8  | 25.9  | 27.2  | (28) |  |
|                                                                        | 2%          | WB          | 24.9  | 24.9  | 24.7  | 22.6  | 20.7  | 18.7  | 18.0  | 19.3  | 20.3  | 21.3  | 23.1  | 24.3  | (29) |  |
|                                                                        |             | MCDB        | 26.8  | 27.0  | 26.5  | 25.1  | 23.1  | 21.2  | 21.0  | 22.0  | 22.6  | 23.3  | 24.8  | 25.9  | (30) |  |
|                                                                        | 5%          | WB          | 24.1  | 24.3  | 23.9  | 22.0  | 20.0  | 17.9  | 17.3  | 18.5  | 19.6  | 20.6  | 22.3  | 23.5  | (31) |  |
|                                                                        |             | MCDB        | 26.0  | 26.4  | 25.8  | 24.5  | 22.7  | 20.7  | 20.3  | 21.2  | 21.7  | 22.6  | 24.0  | 25.2  | (32) |  |
|                                                                        | 10%         | WB          | 23.5  | 23.8  | 23.3  | 21.4  | 19.3  | 17.1  | 16.6  | 17.8  | 18.9  | 20.0  | 21.6  | 22.8  | (33) |  |
|                                                                        |             | MCDB        | 25.5  | 25.9  | 25.3  | 23.8  | 22.1  | 19.9  | 19.7  | 20.5  | 21.0  | 22.0  | 23.3  | 24.7  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.8   | 4.8   | 4.9   | 5.5   | 6.1   | 6.6   | 6.5   | 5.9   | 5.3   | 5.1   | 4.8   | 4.6   | (35) |  |
|                                                                        |             | MCDBR       | 5.7   | 5.5   | 5.7   | 6.5   | 7.7   | 8.2   | 8.5   | 8.0   | 7.0   | 6.5   | 6.2   | 5.7   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 3.3   | 3.0   | 3.3   | 3.9   | 5.0   | 5.6   | 5.8   | 5.2   | 4.4   | 4.0   | 3.9   | 3.4   | (37) |  |
|                                                                        |             | MCWBR       | 5.2   | 5.2   | 5.3   | 6.1   | 6.5   | 6.7   | 6.7   | 6.3   | 6.1   | 5.7   | 5.3   | 5.3   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.441 | 0.436 | 0.424 | 0.391 | 0.367 | 0.349 | 0.359 | 0.422 | 0.465 | 0.468 | 0.444 | 0.443 | (40) |  |
|                                                                        |             | taud        | 2.333 | 2.350 | 2.365 | 2.422 | 2.451 | 2.474 | 2.434 | 2.246 | 2.139 | 2.174 | 2.282 | 2.313 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 903   | 887   | 863   | 840   | 810   | 802   | 809   | 792   | 809   | 848   | 894   | 904   | (42) |  |
|                                                                        |             | Edh at Noon | 136   | 129   | 120   | 102   | 89    | 82    | 89    | 120   | 148   | 153   | 142   | 139   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 5.91  | 5.70  | 4.91  | 4.03  | 3.37  | 3.02  | 3.22  | 3.79  | 4.41  | 4.88  | 5.43  | 5.72  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.51  | 0.43  | 0.26  | 0.27  | 0.18  | 0.15  | 0.21  | 0.24  | 0.29  | 0.47  | 0.41  | 0.42  | (45) |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | (46) |  |
| (47) | Regional (9 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | (47) |  |

Nomenclature: See separate page